

Module - 4

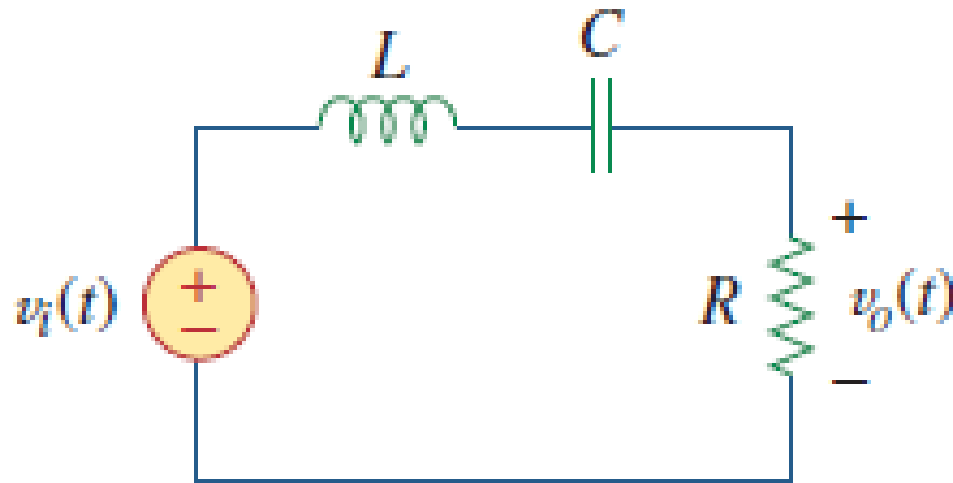
Band Pass and Band Stop Filters

Dr. Sathya P
Associate Professor
SENSE

Band Pass Filter

A **bandpass filter** is designed to pass all frequencies within a band of frequencies, $\omega_1 < \omega < \omega_2$.

The *RLC* series resonant circuit provides a bandpass filter when the output is taken off the resistor



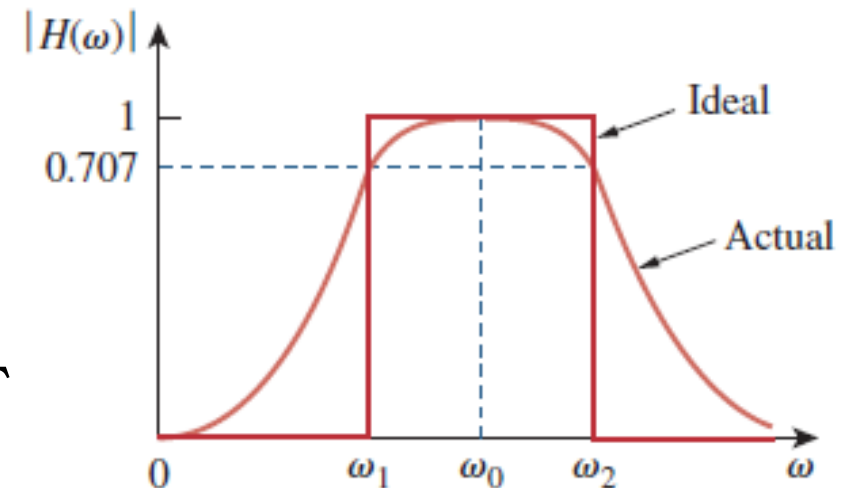
$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{R}{R + j(\omega L - 1/\omega C)}$$

We observe that $\mathbf{H}(0) = 0$, $\mathbf{H}(\infty) = 0$.

The band pass filter passes a band of frequencies ($\omega_1 < \omega < \omega_2$) centered on ω_0 , the center frequency given by

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

- The BPF is a series resonant circuit.
- It can also be formed by cascading HPF and LPF.



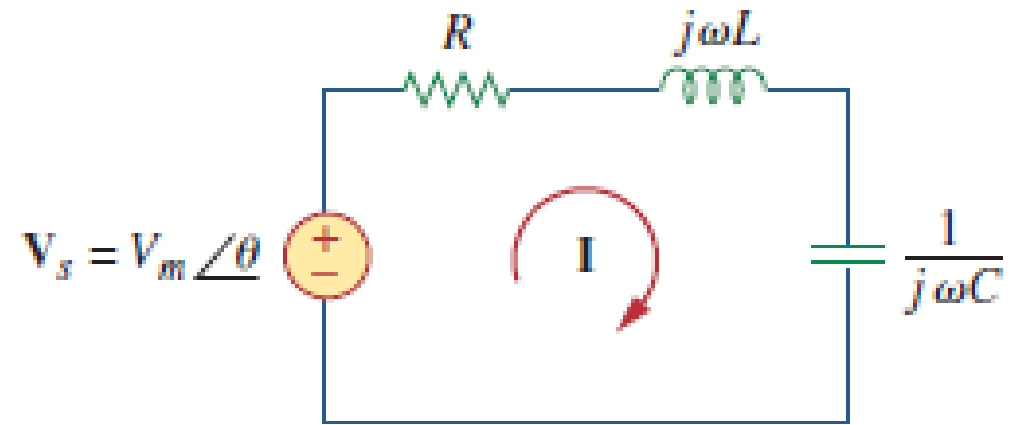
Series Resonant Circuit

- Resonance occurs in any system that has a complex conjugate pair of poles.
- It is the cause of oscillations of stored energy from one form to another.
- It occurs in circuit that has at least one inductor and one capacitor.

Resonance is a condition in an RLC circuit in which the capacitive and inductive reactances are equal in magnitude, thereby resulting in a purely resistive impedance.

$$\mathbf{Z} = \mathbf{H}(\omega) = \frac{\mathbf{V}_s}{\mathbf{I}} = R + j\omega L + \frac{1}{j\omega C}$$

$$\mathbf{Z} = R + j\left(\omega L - \frac{1}{\omega C}\right)$$



Resonance results when the imaginary part of the transfer function is zero, or

$$\text{Im}(\mathbf{Z}) = \omega L - \frac{1}{\omega C} = 0$$

The value of ω that satisfies this condition is called the *resonant frequency* ω_0 . Thus, the resonance condition is

$$\omega_0 L = \frac{1}{\omega_0 C}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/s}$$

Since $\omega_0 = 2\pi f_0$, $f_0 = \frac{1}{2\pi\sqrt{LC}} \text{ Hz}$

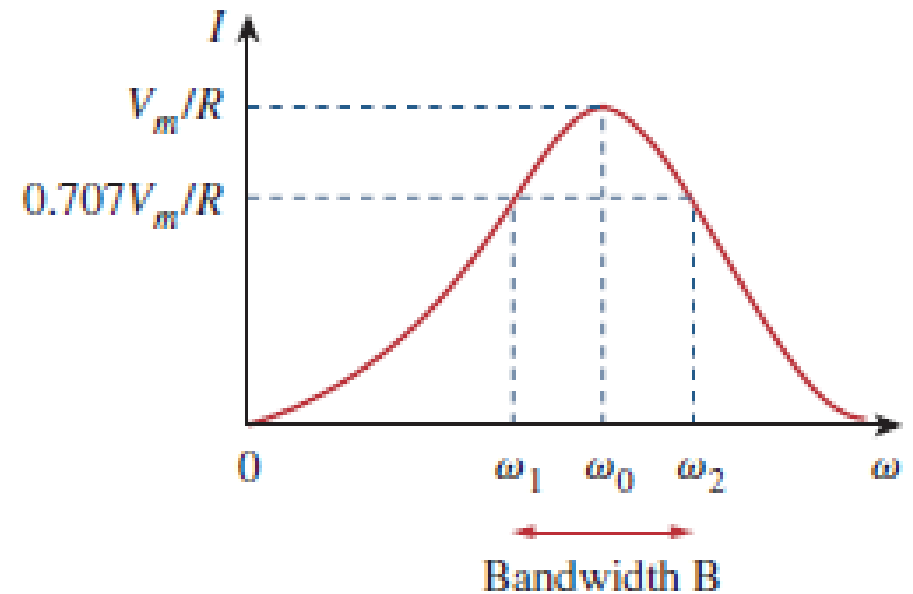
$$I = |\mathbf{I}| = \frac{V_m}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}}$$

$$P(\omega) = \frac{1}{2} I^2 R$$

$$P(\omega_0) = \frac{1}{2} \frac{V_m^2}{R}$$

At certain frequencies $\omega = \omega_1, \omega_2$, the dissipated power is half the maximum value; that is,

$$P(\omega_1) = P(\omega_2) = \frac{(V_m/\sqrt{2})^2}{2R} = \frac{V_m^2}{4R}$$



Hence, ω_1 and ω_2 are called the *half-power frequencies*.

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$\omega_0 = \sqrt{\omega_1 \omega_2}$$

$$B = \omega_2 - \omega_1$$

The “sharpness” of the resonance in a resonant circuit is measured quantitatively by the *quality factor* Q .

$$Q = 2\pi \frac{\text{Peak energy stored in the circuit}}{\text{Energy dissipated by the circuit in one period at resonance}}$$

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R}$$

$$B = \frac{R}{L} = \frac{\omega_0}{Q}$$

$$B = \omega_0^2 CR.$$

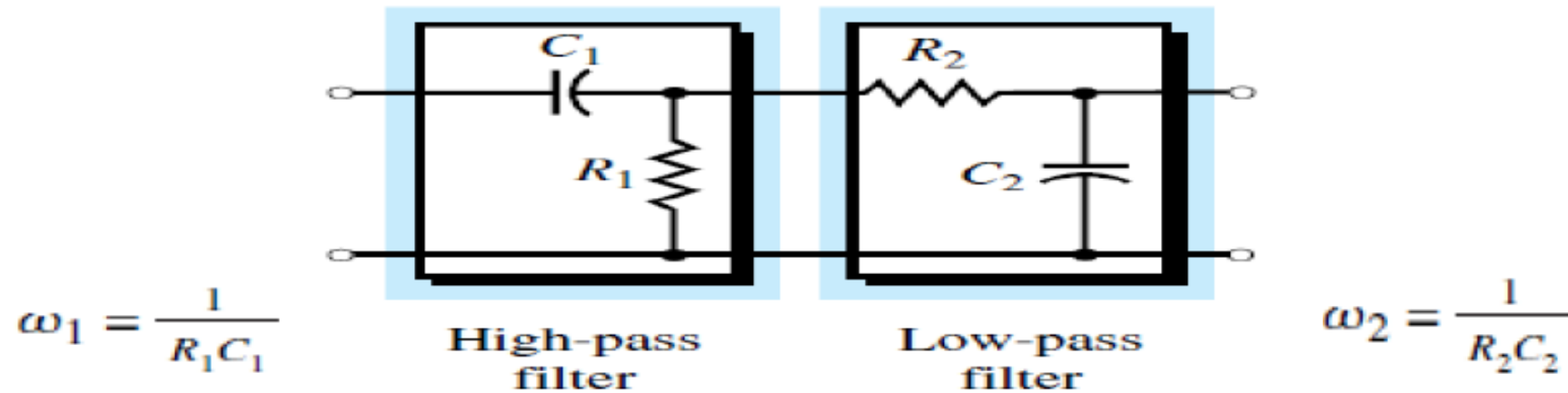
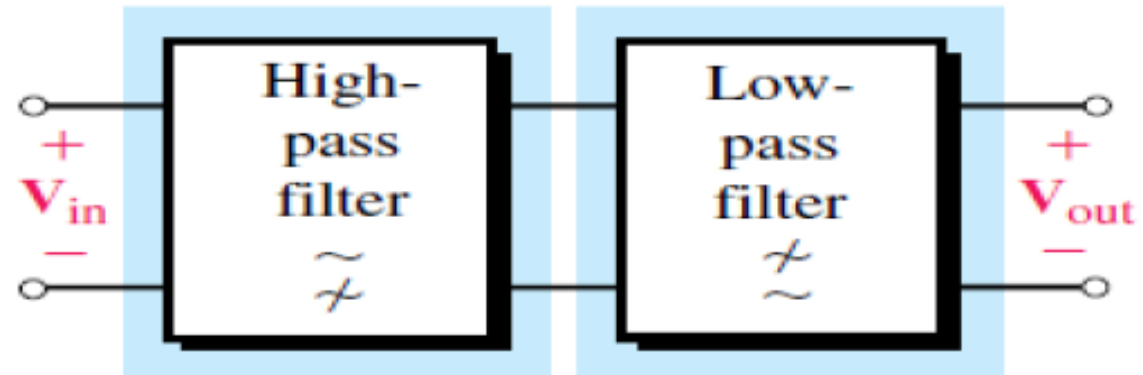
The **quality factor** of a resonant circuit is the ratio of its resonant frequency to its bandwidth.

$$\omega_1 \approx \omega_0 - \frac{B}{2}, \quad \omega_2 \approx \omega_0 + \frac{B}{2}$$

For high- Q circuits ($Q \geq 10$),

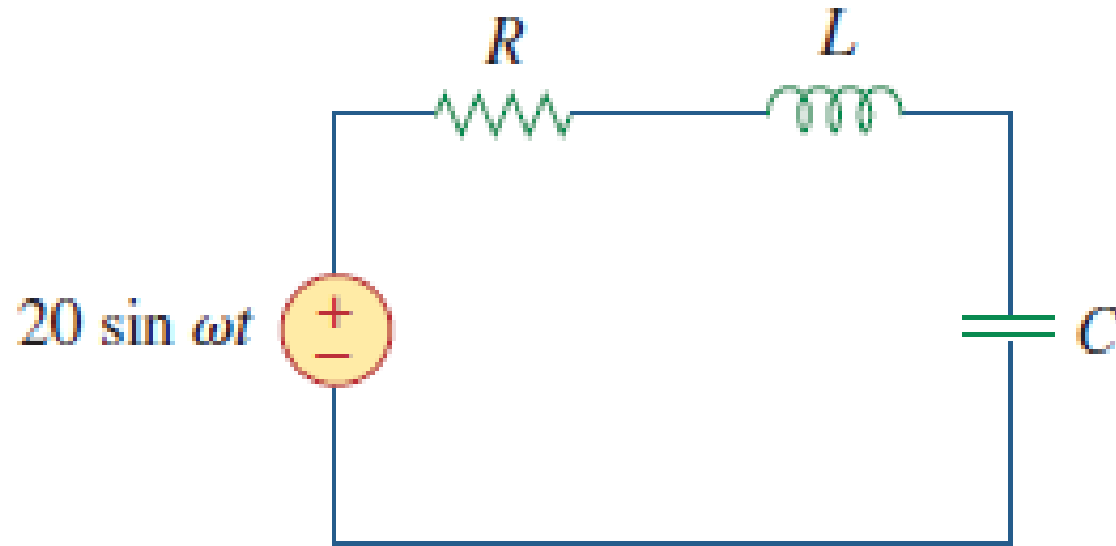
Band Pass filter

- Band pass filter obtained by cascading HPF and LPF.



Numerical - 1

Find the resonant frequency, half power frequencies, quality factor, bandwidth and amplitude of the current at ω_0 , ω_1 , and ω_2 . Given: $R = 2\Omega$, $L = 1 \text{ mH}$, and $C = 0.4 \mu\text{F}$.



Solution:

(a) The resonant frequency is

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-3} \times 0.4 \times 10^{-6}}} = 50 \text{ krad/s}$$

The lower half-power frequency is

$$\begin{aligned}\omega_1 &= -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \\ &= -\frac{2}{2 \times 10^{-3}} + \sqrt{(10^3)^2 + (50 \times 10^3)^2} \\ &= -1 + \sqrt{1 + 2500} \text{ krad/s} = 49 \text{ krad/s}\end{aligned}$$

Similarly, the upper half-power frequency is

$$\omega_2 = 1 + \sqrt{1 + 2500} \text{ krad/s} = 51 \text{ krad/s}$$

(b) The bandwidth is

$$B = \omega_2 - \omega_1 = 2 \text{ krad/s}$$

or

$$B = \frac{R}{L} = \frac{2}{10^{-3}} = 2 \text{ krad/s}$$

The quality factor is

$$Q = \frac{\omega_0}{B} = \frac{50}{2} = 25$$

(c) At $\omega = \omega_0$,

$$I = \frac{V_m}{R} = \frac{20}{2} = 10 \text{ A}$$

At $\omega = \omega_1, \omega_2$,

$$I = \frac{V_m}{\sqrt{2}R} = \frac{10}{\sqrt{2}} = 7.071 \text{ A}$$

■ **METHOD 2** Alternatively, we could find

$$Q = \frac{\omega_0 L}{R} = \frac{50 \times 10^3 \times 10^{-3}}{2} = 25$$

From Q , we find

$$B = \frac{\omega_0}{Q} = \frac{50 \times 10^3}{25} = 2 \text{ krad/s}$$

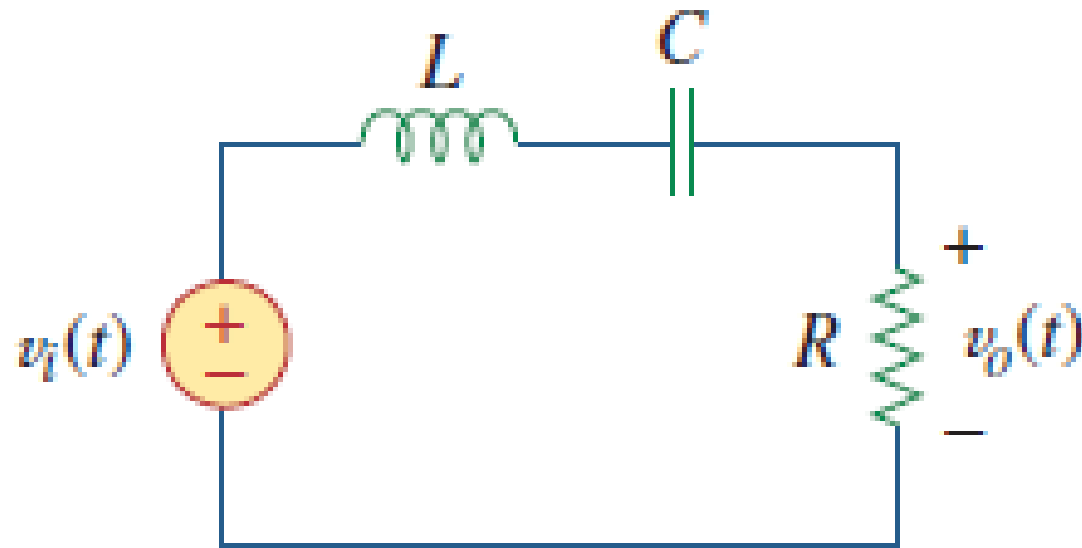
Since $Q > 10$, this is a high- Q circuit and we can obtain the half-power frequencies as

$$\omega_1 = \omega_0 - \frac{B}{2} = 50 - 1 = 49 \text{ krad/s}$$

$$\omega_2 = \omega_0 + \frac{B}{2} = 50 + 1 = 51 \text{ krad/s}$$

Numerical - 2

Design a bandpass filter of the form shown below with a lower cutoff frequency of 20.1 kHz and an upper cutoff frequency of 20.3 kHz. Take $R = 20 \text{ k}\Omega$. Calculate L , C , and Q .



Answer: 15.915 H, 3.9 pF, 101.

Numerical - 3

Design a series RLC type bandpass filter with cutoff frequencies of 10 kHz and 11 kHz. Assuming $C = 80 \text{ pF}$, find R , L , and Q .

Solution:

$$\omega_1 = 2\pi f_1 = 20\pi \times 10^3$$

$$\omega_2 = 2\pi f_2 = 22\pi \times 10^3$$

$$B = \omega_2 - \omega_1 = 2\pi \times 10^3$$

$$\omega_0 = \frac{\omega_2 + \omega_1}{2} = 21\pi \times 10^3$$

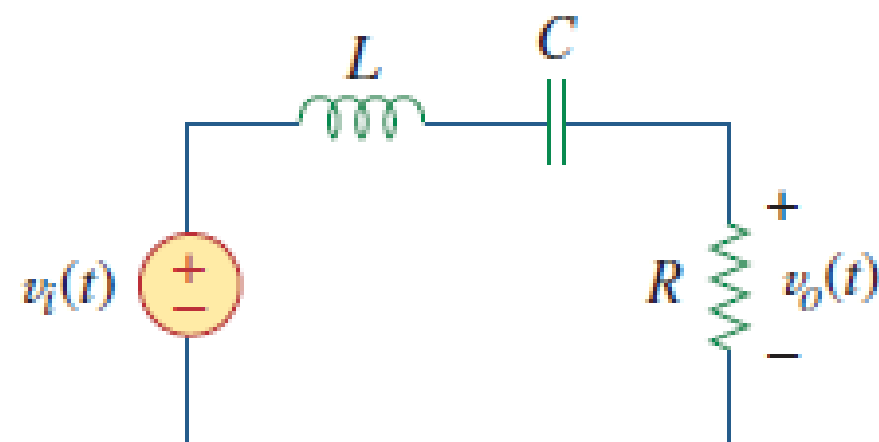
$$Q = \frac{\omega_0}{B} = \frac{21\pi}{2\pi} = \underline{\underline{10.5}}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \longrightarrow L = \frac{1}{\omega_0^2 C}$$

$$L = \frac{1}{(21\pi \times 10^3)^2 (80 \times 10^{-12})} = \underline{\underline{2.872 \text{ H}}}$$

$$B = \frac{R}{L} \longrightarrow R = BL$$

$$R = (2\pi \times 10^3)(2.872) = \underline{\underline{18.045 \text{ k}\Omega}}$$



Numerical - 4

Determine the range of frequencies that will be passed by a series RLC type bandpass with $R = 10 \, \Omega$, $L = 25 \, \text{mH}$, and $C = 0.4 \, \mu\text{F}$. Find the quality factor.

Solution:

$$\omega_o = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(25 \times 10^{-3})(0.4 \times 10^{-6})}} = 10 \, \text{krad/s}$$

$$B = \frac{R}{L} = \frac{10}{25 \times 10^{-3}} = 0.4 \, \text{krad/s}$$

$$Q = \frac{10}{0.4} = \underline{25}$$

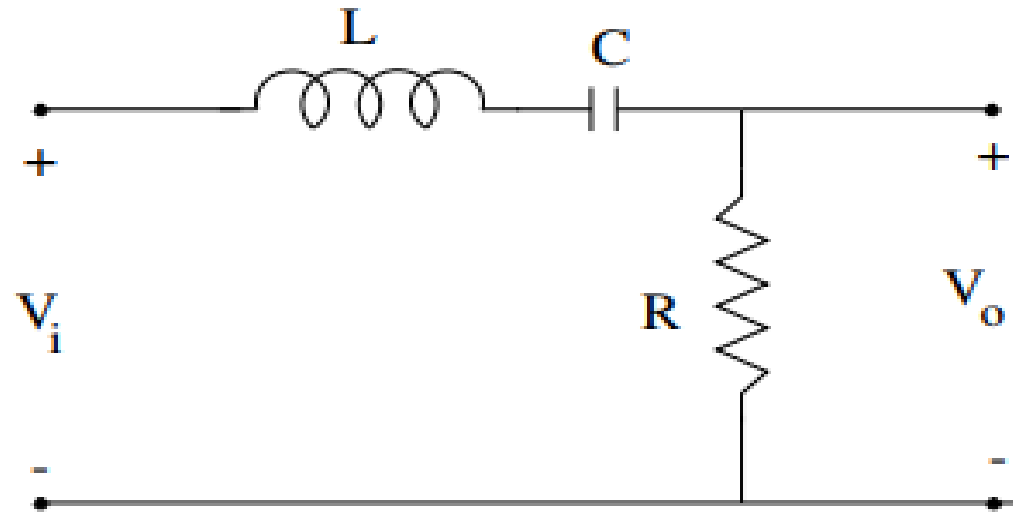
$$\omega_1 = \omega_o - B/2 = 10 - 0.2 = 9.8 \, \text{krad/s} \quad \text{or} \quad f_1 = \frac{9.8}{2\pi} = 1.56 \, \text{kHz}$$

$$\omega_2 = \omega_o + B/2 = 10 + 0.2 = 10.2 \, \text{krad/s} \quad \text{or} \quad f_2 = \frac{10.2}{2\pi} = 1.62 \, \text{kHz}$$

$$\underline{1.56 \, \text{kHz} < f < 1.62 \, \text{kHz}}$$

Numerical - 5

Design a RLC bandpass filter with a lower cut-off frequency of 1 kHz and a bandwidth of 3 kHz. What is the center frequency and Q of this filter?



- Solution:

$$B(\text{Hz}) = f_u - f_l \qquad f_u = 1 + 3 = 4 \text{ kHz}$$

$$B(\text{rad/s}) = 2\pi B(\text{Hz}) = 1.88 \times 10^4$$

$$\omega_u = 2\pi f_u = 2.51 \times 10^4 \qquad \omega_l = 2\pi f_l = 6.28 \times 10^3$$

$$\omega_0 = \sqrt{\omega_u \omega_l} = 1.26 \times 10^4$$

$$B(\text{rad/s}) = \frac{\omega_0}{Q} \quad \rightarrow \quad Q = \frac{1.26 \times 10^4}{1.88 \times 10^4} = 0.67$$

For the series RLC circuit:

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad \rightarrow \quad LC = \frac{1}{\omega_0^2}$$

$$C = \frac{1}{L\omega_0^2} = \frac{1}{10 \times 10^{-3} \times (1.26 \times 10^4)^2} = 0.63 \text{ } \mu\text{F}$$

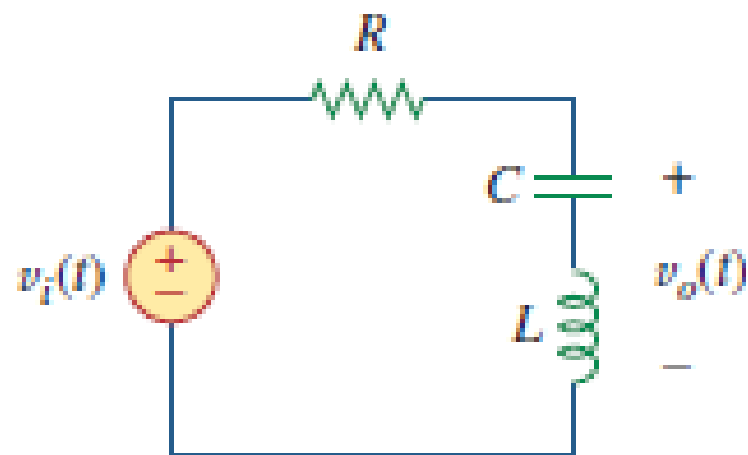
$$Q = \frac{\omega_0}{R/L} \quad \rightarrow \quad \frac{R}{L} = \frac{\omega_0}{Q} = B(\text{rad/s})$$

$$R = LB = 10 \times 10^{-3} \times 1.88 \times 10^4 = 188 \text{ } \Omega$$

Band Stop Filter

A bandstop filter is designed to stop or eliminate all frequencies within a band of frequencies, $\omega_1 < \omega < \omega_2$.

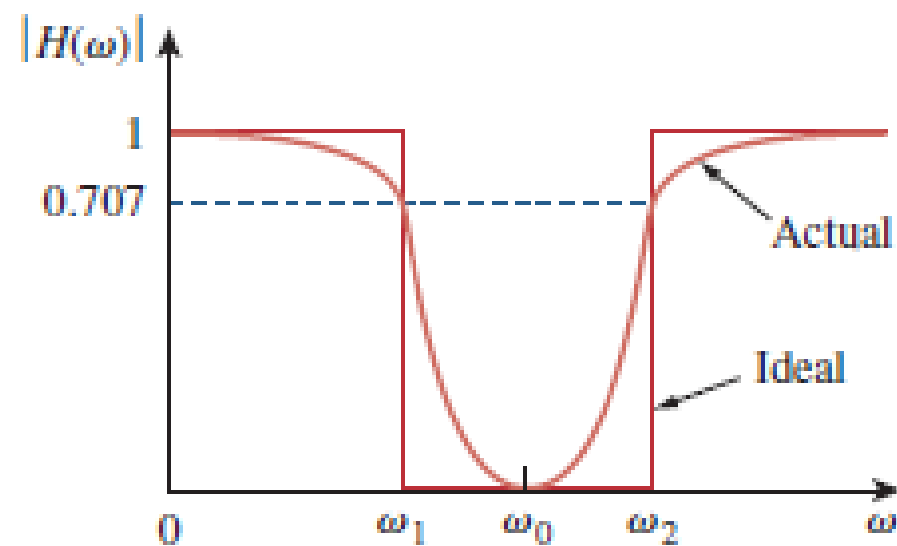
- A filter that prevents a band of frequencies between two designated values (ω_1 and ω_2) from passing is variably known as a band stop, band-reject or notch filter.
- A bandstop filter is formed when the output RLC series resonant circuit is taken off the LC series.



$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{j(\omega L - 1/\omega C)}{R + j(\omega L - 1/\omega C)}$$

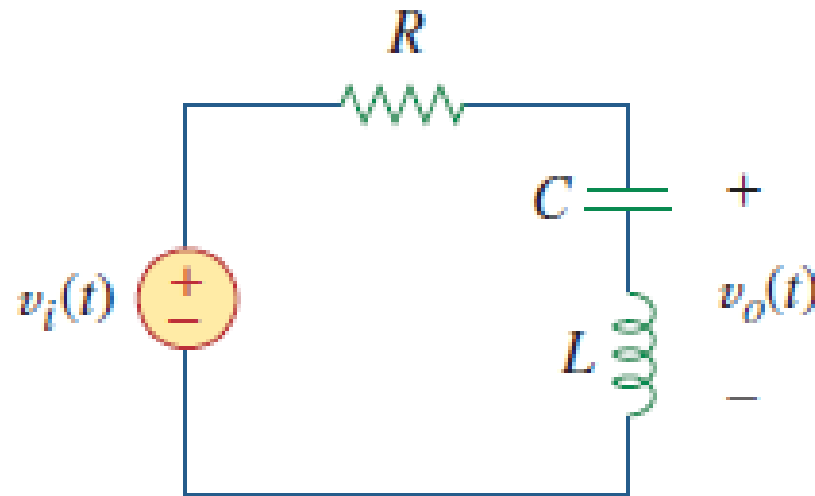
$$\mathbf{H}(0) = 1, \quad \mathbf{H}(\infty) = 1.$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$



Numerical - 1

- If the bandstop filter in Figure below is to reject a 200 Hz sinusoid while passing other frequencies, calculate the values of L and C . Take $R = 150\ \Omega$ and the bandwidth as 100 Hz.



Solution:

We use the formulas for a series resonant circuit

$$B = 2\pi(100) = 200\pi \text{ rad/s}$$

$$B = \frac{R}{L} \quad \Rightarrow \quad L = \frac{R}{B} = \frac{150}{200\pi} = 0.2387 \text{ H}$$

Rejection of the 200-Hz sinusoid means that f_0 is 200 Hz,

$$\omega_0 = 2\pi f_0 = 2\pi(200) = 400\pi$$

Since $\omega_0 = 1/\sqrt{LC}$,

$$C = \frac{1}{\omega_0^2 L} = \frac{1}{(400\pi)^2(0.2387)} = 2.653 \mu\text{F}$$

Numerical - 2

- The circuit parameters for a series RLC bandstop filter are $R = 2 \text{ k}\Omega$, $L = 0.1 \text{ H}$, and $C = 40 \text{ pF}$. Calculate the center frequency, the half power frequencies and the quality factor.

- Solution:

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(0.1)(40 \times 10^{-12})}} = \underline{0.5 \times 10^6 \text{ rad/s}}$$

$$B = \frac{R}{L} = \frac{2 \times 10^3}{0.1} = 2 \times 10^4 \quad Q = \frac{\omega_0}{B} = \frac{0.5 \times 10^6}{2 \times 10^4} = 25$$

As a high Q circuit,

$$\omega_1 = \omega_0 - \frac{B}{2} = 10^4 (50 - 1) = \underline{490 \text{ krad/s}}$$

$$\omega_2 = \omega_0 + \frac{B}{2} = 10^4 (50 + 1) = \underline{510 \text{ krad/s}}$$